PROGRAM PROPOSAL
Home Electricity Cost Estimator

Name: Catu, Ryan David C.
Section: BSIT-IT301-3B
Program Title: Home Electricity Cost Estimator
Programming Language: C#

I. Description and Purpose

The Home Electricity Cost Estimator is a C# console application that estimates the electricity consumption and cost of household appliances. The user enters the appliance name, wattage, and number of hours it is used per day. The program calculates the estimated electricity consumption in kilowatt-hours (kWh) and the cost based on the billing period and electricity rate. It also compares the estimated cost with a monthly budget of ₱8,000. The purpose of this project is to practice basic C# programming concepts while creating a simple tool that can help users estimate their household electricity expenses.

II. Inputs

- Billing period in days (1–31)
- Appliance name
- Appliance wattage (1–10,000 watts)
- Daily operating hours (0.01–24 hours)
- Electricity rate per kWh
- Monthly electricity budget
- Menu selection

III. Processes

1. Display the program title and request the billing period.
2. Show the main menu and accept the user's choice.
3. Validate the user's input to ensure that it meets the allowed requirements.
4. Store appliance details in arrays.
5. Calculate daily electricity consumption using:
   Daily kWh = (Wattage × Hours Used) ÷ 1,000
6. Calculate billing-period consumption using:
   Total kWh = Daily kWh × Billing Days
7. Calculate the estimated cost using:
   Estimated Cost = Total kWh × Electricity Rate
8. Compare the estimated cost with the monthly budget.
9. Display the results and repeat the menu until the user chooses Exit.
10. Handle unexpected exceptions and end the program properly.

IV. Outputs

- List of saved appliances
- Daily electricity consumption per appliance
- Estimated electricity consumption for the billing period
- Estimated electricity cost per appliance and total cost
- Budget status and remaining budget, when applicable
- Input validation messages and error messages
- Program termination message

V. Pseudocode

```
START
    Display program title
    Input and validate billing days

    REPEAT
        Display main menu
        Input and validate menu choice

        SWITCH menu choice
            CASE 1: Add appliance
            CASE 2: View appliances
            CASE 3: Calculate electricity cost
            CASE 4–9: Features to be implemented
            CASE 10: Exit program
        END SWITCH

    UNTIL exit is selected

    Handle unexpected exceptions
    Display program termination message
END
```

VI. Expected Result

The program is expected to accept valid appliance information, calculate estimated electricity consumption and costs, compare the estimated total with the monthly budget, and provide understandable results. It also demonstrates the use of variables, constants, arrays, conditional statements, loops, operators, input validation, and exception handling.